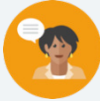


11.6 Writing Exponential Functions From a Written Description

Lesson Introduction

 Benchmarks	MA.912.AR.5.3 Given a mathematical or real-world context, classify exponential functions as representing growth or decay. MA.912.AR.5.4 Write an exponential function to represent a relationship between two quantities from a graph, a written description, or a table of values within a mathematical or real-world context.		 Learning Targets	- I can write an exponential function from a written description. - I can classify a real-world context modeled by an exponential function as exponential growth or decay.
 Benchmark Coherence	Building On		Working Toward	
	N/A		MA.912.AR.5.5: Given an expression or equation representing an exponential function, reveal the constant percent rate of change per unit interval using the properties of exponents. Interpret the constant percent rate of change in terms of a real-world context.	
 Essential Question	How can you write an exponential function that represents a real-world context?			
 Lesson Narrative	In this lesson, students extend their learning about writing exponential functions to include real-world contexts. Students begin by learning how to write exponential functions given a written description using two real-world scenarios. They will then practice two problems on their own before transitioning into a Collaboration activity. Lastly, students will try three real-world problems on their own and complete a self-reflection.			
 Component Summary	11.6.1 Warm-Up (~5 min.)	Math Talk activity (<i>SE p. 186</i>)	11.6.4 Collaboration (10 min.)	Collaborative activity writing exponential functions from a written description (<i>SE p. 189</i>)
	11.6.2 Guided Instruction (10 min.)	Introduction to writing exponential functions from a written description (<i>SE p. 187</i>)	11.6.5 Your Turn! (5-10 min.)	Independent practice writing exponential functions from a written description (<i>SE p. 190</i>)
	11.6.3 Try It (10 min.)	Practice writing exponential functions from a written description (<i>SE p. 188</i>)	11.6.6 Wrap-Up (~5 min.)	Emoji self-reflection on the learning targets from the unit (<i>SE p. 191</i>)
 Resources	Glossary		Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> N/A		Work through each of these problems prior to facilitating with students to determine contexts or language that students may need additional support with.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle to determine the correct values of a and b from a written description. - Students may misinterpret a decay value from the written description (i.e., loses 10% being written as 0.10 instead of 0.9 in the function).	If time is an issue, consider assigning 11.6.5 Your Turn! as homework.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Three Reads This lesson consists primarily of word problems and contexts that require the student to fully understand what is occurring in the problem. Have students engage in this protocol to ensure students are aware of the full context prior to attempting to answer questions about the context.	MA.K12.MTR.7.1: Real-World Contexts This lesson consists of a collection of problems where exponential functions need to be graphed based on a written description and equation that are based in a real-world context. Students will use models and methods to understand, represent, and solve problems throughout this lesson.
 Differentiate Instruction	Since this lesson requires a greater degree of reading comprehension, consider grouping your students for the 11.6.4 Collaboration by reading level in order to work with them in smaller groups at the stages in the process where they may need your assistance the most. Student performance during 11.6.5 Your Turn! can provide insight into how well students are understanding how to write an exponential function from a written description.	
Lesson Notes:		

11.6.1 Warm-Up: Math Talk

Component Overview: Students will complete a Math Talk activity where they compare the size of South Carolina and Florida from a map.



11.6.1 Warm-Up: Math Talk

The map shows the Southeastern United States to scale. Using the map, fill in the blanks below. Be prepared to share your reasoning with the class.



1. Florida has an area that is about _____ times bigger than South Carolina.

Answers will vary.

2. The area of Florida is 65,755 square miles. Based on your estimate, the area of South Carolina is _____ square miles.

Answers will vary.



When reviewing this activity, consider having students share out their answers for both questions. Chart up the various answers that come out of that process. Then have students vote on whether they believe each charted answer is reasonable. Conclude by getting the actual answer for the area of South Carolina and allow students a chance to determine who had the most accurate estimate.

11.6.2 Guided Instruction: Writing Exponential Functions

Component Overview: Students will be introduced to writing exponential functions given a written description. In this activity, there will be two real-world problems to guide students through this process.



11.6.2 Guided Instruction: Writing Exponential Functions

1. A typical cup of coffee contains about 100 milligrams (mg) of caffeine. Every hour approximately 6% of the amount of caffeine in the body is metabolized and eliminated.



- a. Discuss with your partner whether the situation represents exponential growth or exponential decay. Justify your choice.
Answers will vary. This represents decay because you start with 100 milligrams of caffeine, and then it starts to eliminate, or decay, every hour.

- b. What is the rate of growth or the rate of decay?
6%

- c. What is the initial amount of caffeine consumed in this scenario?
100 milligrams

- d. Let C represent the amount of caffeine in the body, in mg, and t represent the number of hours since a cup of coffee was consumed. Write C as a function of t .
 $C(t) = 100(0.94)^t$



Students may need to refer to forms of exponential functions introduced in Lesson 5. It may be beneficial to have students recall what that form is and what each value means. This can help connect the guided questions used during question 1 to the final answer in question 1d.



Why would the function not use 6 as the value of b ?

11.6.2 Guided Instruction: Writing Exponential Functions (continued)

2. In 2020, the population of Florida was estimated to be 21.7 million people, with an annual rate of increase of 1.12%.

- a. Explain whether the population can be modeled by exponential growth or exponential decay.

This would be exponential growth because it is increasing.

- b. Wyatt and Carter both wrote a function to model Florida's population growth, where x is the number of years since 2020, and P is the population in millions. Both students made a mistake. Circle the error each student made.

Wyatt	Carter
$P(x) = 21.7(1.12)^x$	$P(x) = 21.7(1 + 0.112)^x$

- c. Discuss with your partner the errors Wyatt and Carter made.

Both students represented their percentages incorrectly.

- d. Write the correct function that models the context using both forms.

$P(x) = a(b)^x$	$P(x) = a(1 + r)^x$
$P(x) = 21.7(1.0112)^x$	$P(x) = 21.7(1 + 0.0112)^x$



For question 2, consider leveraging a Think-Pair-Share protocol. Have students work on questions 2a and 2b on their own. After they have completed those, allow them to work with a partner to complete questions 2c and 2d. When all partner pairs have finished or when time is called, have students share their answers to each part of this question.



- How would you explain to Wyatt what he did wrong and help prevent him from making that same mistake again?*
- How would you explain to Carter what he did wrong and help prevent him from making that same mistake again?*

11.6.3 Try It: Writing Exponential Functions

Component Overview: Students will try two questions of writing exponential functions given a written description on their own and then with a partner.



11.6.3 Try It: Writing Exponential Functions

Work independently to write the function for the two scenarios.

1. The value of a 2021 Ford Bronco was \$44,205 when it went on the market. It depreciates at a rate of 10.5% per year.



- a. Is this an example of growth or decay?

Decay

- b. Write a function that models the car's value, where V is the value of car and t is the number of years since 2021.

$$V(t) = 44,205(.895)^t$$

2. The Florida Fish and Wildlife Conservation Commission have found that the Florida black bear population is increasing due to the Florida Black Bear Management Plan. In 2019, there were 4,050 bears in Florida with a rate of increase of 6.5% per year.



- a. Is this an example of growth or decay?

Growth

- b. Write an equation that models the population, where B is the number of black bears and x is the years since 2019.

$$B(x) = 4,050(1.065)^x$$

3. Compare your answers to questions 1-2 with your partner, and if necessary, resolve any differences.

Answers will vary.



For this activity, allow students to complete both questions 1 and 2 independently in order to practice what they've learned. In question 3, students can work with their partner to compare their answers and rationale in order to come to a consensus on the correct answer.



How can you tell based on the scenario that this will be modeled by a decay function?



How can you determine that this is an example of exponential growth?

11.6.4 Collaboration: Exponential Equations in Real-World Contexts

Component Overview: Students will work collaboratively with their partner to explore the exponential function of a given scenario involving the regulation of sale.



11.6.4 Collaboration: Exponential Equations in Real-World Contexts

U.S. lawmakers have tried to pass legislation that would reduce the consumption of salt. Versions of this law have been proposed in the past that would mandate restaurants to decrease sodium levels in the food they serve by an exponential rate. The sodium level in food is the amount of salt in the food.

It is estimated that all U.S. restaurants use a collective total of 5 million grams of salt a year. If the law were to pass, they would be required to reduce to 4.875 million grams the following year and continue reducing at a constant percentage each year until a goal is reached.

1. Discuss with your partner how the values given in this scenario can be used to write an exponential function that models the situation.
5 million would be the initial value and 0.975 would be the decay factor.
2. Two students discussed how to determine the value of b to write an exponential function in the form $y = a(b)^x$. Their thinking is shown in the table.

Jayden	Ella
"I would subtract 4.875 million from 5 million to find the rate of change from one year to the next."	"I would divide 4.875 million by 5 million to find the common ratio from one year to the next."

Explain which student is correct.

Ella is correct because this is an exponential decay, not a linear decrease.



Consider leveraging a Three Reads strategy with this activity prior to having students begin working on the questions with their partner. This can help support reading comprehension and meta-awareness of mathematical language. It also supports negotiating information in the text with their partner in mathematical conversation.



- What was the error in Jayden's thinking?
- How would you illustrate for Jayden his misconception and how to keep it from happening in the future?

11.6.4 Collaboration: Graphing Exponential Functions (continued)

3. Find the value of b in this scenario.

0.975

4. Discuss with your partner which value in the scenario represents a .

5 million

5. Work with your partner to write the function rules in the forms indicated, where S is the total amount of salt that can be used, in millions of grams, for the given years, x , since the law passed.

$S(x) = a(b)^x$	$S(x) = a(1 \pm r)^x$
$S(x) = 5(.975)^x$	$S(x) = 5(1 - .025)^x$

6. What is the constant percent rate of change by which the law would require restaurants to reduce the amount of salt served in their food each year?

97.5%



What indicators or language from the problem helped you locate the values of a and b ?



Students may make an error when completing the right-hand side of the column. Some students will write $(1 - 0.975)$. By asking students with this error what it means or how it relates to the context, they will often catch their own error in answering the question.

11.6.5 Your Turn!

Component Overview: Students will practice writing exponential functions from a written description for three real-world contexts.



11.6.5 Your Turn!

1. In 1987, there were 17 Starbucks locations. The number of shops has increased 29% annually since then.

Write the function that models the number of Starbucks since 1987. Let t be the number of years since 1987 and S be the number of shops.

$$S(t) = 17(1.29)^t$$

2. There were 1.462 million paid subscriptions for streaming music in 2010. Since then, there has been a 54.4% per year increase in paid music streaming subscriptions.

Write a function that represents the number of paid subscriptions since 2010. Let x be the years since 2010 and P be the number of paid music streaming subscriptions, in millions.

$$P(x) = 1.462(1.544)^x$$

3. An adult takes 400 milligrams (mg) of ibuprofen. Each hour, the amount of ibuprofen in the person's system decreases by 29%.

Write a function that represents the amount of ibuprofen, A , in mg, after h hours.

$$A(h) = 400(0.71)^h$$



- How does knowing if the situation models growth or decay help you determine what the equation will look like?
- How can you determine the initial value from the context?



Use the time during this activity to determine which students may need additional support and consider pulling small groups together to provide additional differentiated support based on where they are struggling.

11.6.6 Wrap-Up: Reflection

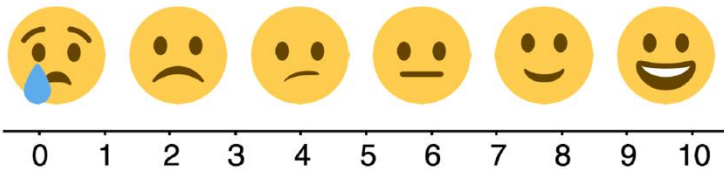
Component Overview: Students will complete an emoji self-reflection on their level of understanding for each of the learning targets from this unit.



11.6.6 Wrap-Up: Reflection

A scale is shown with emojis that correlate to different levels of confidence.

Answers will vary.



1. Draw the emoji that best describes your understanding of each learning target from this unit.

Emoji	Learning Target
	I can classify a function as representing exponential growth or decay from a graph or table.
	I can determine the key features of an exponential function from a table, graph, or exponential function.
	I can graph an exponential function given the equation.
	I can graph an exponential function given a written description and determine its key features.
	I can write an exponential function from a graph or table.
	I can write an exponential function from a written description.



What would reflection look like for you as a teacher? Consider your own informal data regarding students' understanding of these learning targets throughout the unit and how to best to address potential gaps in understanding prior to the unit assessment.

Consider making notes on the unit guide for things you would want to keep in mind for this unit next school year.